

A Randomization-Control Diagnostic for Document Grounding in Self-Distillation Gates

Xiankun Lin^{1,*} Lixin Liu^{2,*} Xinyi Lin^{1,*} Zhezheng Hao^{3,†}

¹Sun Yat-sen University, Guangzhou, China

²Tsinghua University, Beijing, China ³Zhejiang University, Hangzhou, China

{linxk7,linxy239}@mail2.sysu.edu.cn liulx25@mails.tsinghua.edu.cn

haozhezheng@outlook.com

*Equal contribution. †Corresponding author.

Abstract

Self-distillation for grounded question answering often filters tutor trajectories with agreement or confidence gates. For such gates to support faithful grounding, their signals should depend on the supplied document rather than on answers recoverable from parametric memory. We propose a four-mode randomization-control diagnostic that applies the same frozen gate to original, randomized, irrelevant, and empty documents for the same question. On HotpotQA and MuSiQue, using Qwen2.5-7B-Instruct with eight tutor traces, the joint gate changes sharply with document condition: kept fraction falls from 0.380 to 0.020 and from 0.330 to 0.015, respectively. Randomized and irrelevant documents lie between the endpoints, while a document-blind random gate stays flat. The protocol audits document sensitivity without additional student training; faithful evidence use and downstream gains require separate validation.

1 Introduction

Grounded language-model systems are only as faithful as the signals used to decide when the model has actually used its evidence. In self-distillation with privileged context, this decision is made by a gate: a tutor sees a document and generates trajectories, while a later student may not see that document; the gate keeps trajectories when sampled answers agree, optionally with confidence (Stein et al., 2026; Taubenfeld et al., 2025). The missing check is whether this reliability signal is itself grounded. A gate can fire because the document supports a stable answer, but it can also fire because the model already knows the answer from parametric memory, especially on QA benchmarks containing memorized facts (Yang et al., 2018; Trivedi et al., 2022; Mallen et al., 2023).

We propose a direct diagnostic. For the same question, we evaluate the tutor under four document modes: original matched, same-benchmark

random, off-domain irrelevant, and empty. We then apply the same frozen gate, with fixed thresholds shared across document modes. We test whether retention changes across document conditions; a document-independent random gate provides a flat control (Figure 1). We instantiate the diagnostic on HotpotQA and MuSiQue. Across both, kept fraction falls sharply from original to no-document inputs, corrupted documents lie between the extremes, confidence-only/agreement-only/joint gates agree, and a fixed-probability random gate stays flat. The result is a reusable document-sensitivity diagnostic for training-time gate signals rather than a new QA model.

2 Related Work

Self-distillation gates. Knowledge distillation transfers supervision from a stronger or better-informed teacher to a student (Hinton et al., 2015; Kim and Rush, 2016; Vapnik and Izmailov, 2015; Lopez-Paz et al., 2015). In LLM reasoning, self-consistency treats agreement across sampled traces as evidence of answer reliability (Wang et al., 2022; Wei et al., 2022). GATES filters document-grounded tutor trajectories under privileged context (Stein et al., 2026), and related work augments agreement with confidence (Taubenfeld et al., 2025). These works define filtering rules, but not whether the gate reads the document.

Document grounding and leakage. Retrieval-augmented and document-grounded QA condition generation on retrieved or supplied passages (Lewis et al., 2020; Guu et al., 2020; Karpukhin et al., 2020; Asai et al., 2024); multi-hop benchmarks stress evidence combination (Yang et al., 2018; Ho et al., 2020; Trivedi et al., 2022). Yet models may ignore evidence (Liu et al., 2024), and context-memory conflicts show parametric knowledge overriding text (Longpre et al., 2021; Xu et al., 2024). Leakage-resistant benchmarks and ground-

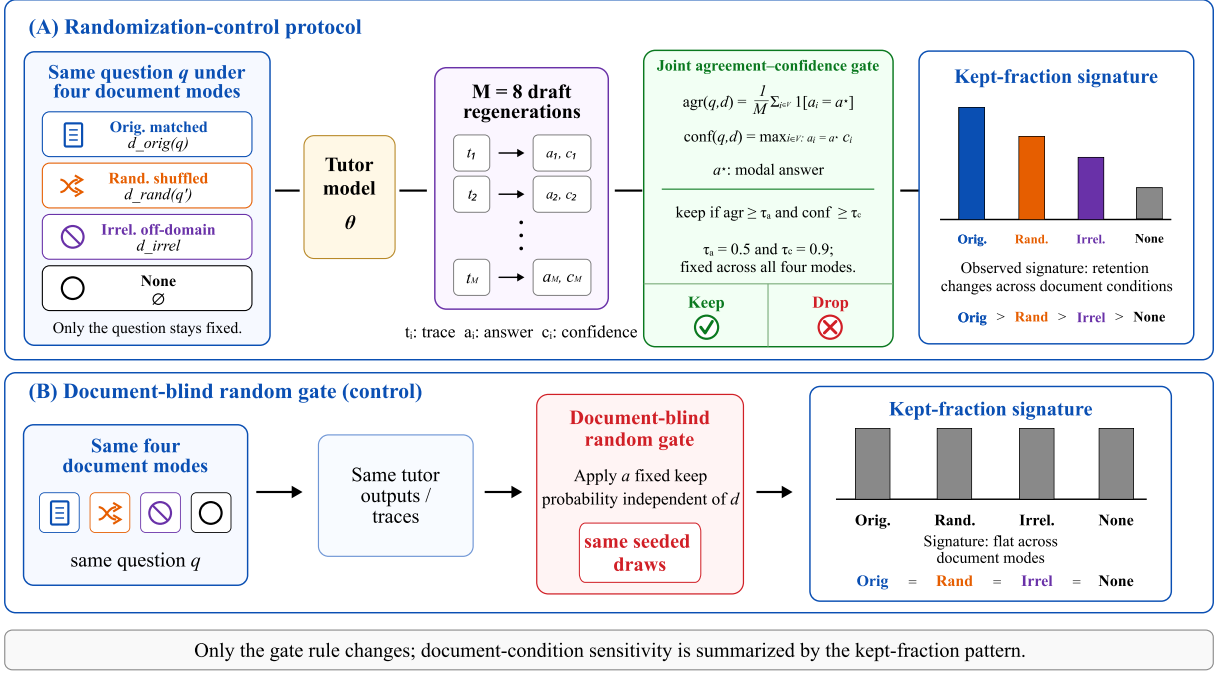


Figure 1: Randomization-control diagnostic. A: apply the same agreement–confidence rule across four document conditions. B: replace it with a document-independent fixed-probability random rule. Signature bars illustrate the expected contrast; measured retention appears in Table 1.

edness diagnostics address inference-time answers (Trivedi et al., 2022; Wu et al., 2025; Dziri et al., 2022; Manakul et al., 2023; Zhang et al., 2025; Kuhn et al., 2023; Vankov et al., 2025); we apply the same concern to training-time gates.

3 Diagnostic

Gate signal. For a question q and document d , the tutor samples M traces. Let V index parsed answers and a^* be their modal answer. We compute $\text{agr}(q, d) = M^{-1} \sum_{i \in V} \mathbf{1}[a_i = a^*]$ and $\text{conf}(q, d) = \max_{i \in V: a_i = a^*} c_i$. Here c_i averages the maximum next-token softmax probability over the generated continuation; it is not restricted to the extracted answer span. If V is empty, both signals are zero. Otherwise agreement is at least $1/M$; means below $1/M$ can arise only when averaging in all-unparseable cases. The joint gate requires $\text{agr} \geq \tau_a$ and $\text{conf} \geq \tau_c$. The released configuration fixes $\tau_a = 0.5$ and $\tau_c = 0.9$ across document modes.

Document modes and signature. For each question, *original* uses the matched document; *randomized* uses a different same-benchmark document; *irrelevant* uses a cross-benchmark document; and *none* provides an empty document. We report kept fraction K and invariance spread

$\Delta K = K_{\text{orig}} - \min(K_{\text{rand}}, K_{\text{irr}}, K_{\text{none}})$. The observed ordering original > randomized > irrelevant > none describes this study; it is not a requirement that every grounded gate must satisfy. Unmatched documents may still contain useful information or distract the tutor. We contrast HotpotQA with MuSiQue’s leakage-resistant construction (Trivedi et al., 2022; Wu et al., 2025).

4 Experimental Setup

Tutor and traces. We use Qwen2.5-7B-Instruct (Qwen et al., 2025) with $M = 8$ traces at temperature 0.8 and 200 two-hop questions per benchmark. Main results use seeds 0 and 1. Irrelevant documents are sampled cross-benchmark. Gate signals are collected before student updates. The archived runner then performs document-conditioned self-distillation on retained questions and evaluates the updated model on 200 validation questions at temperature zero. Doc Acc. supplies original matched documents at evaluation; No-doc Acc. omits them on the same validation split. Thus the row’s document mode describes signal collection and training, not evaluation. Main-table entries are mean $\pm$ sample standard deviation across seeds.

Benchmark	Doc. mode	Doc Acc.	No-doc Acc.	Keep	Agree	Conf.	Correct
HotpotQA	Original	.320±.014	.145±.035	.380±.021	.479±.005	.902±.002	.287±.011
HotpotQA	Randomized	.273±.011	.165±.028	.225±.014	.422±.011	.845±.006	.323±.011
HotpotQA	None	.300±.007	.055±.000	.020±.000	.121±.002	.225±.002	.117±.004
HotpotQA	Irrelevant	.255±.021	.143±.018	.098±.004	.356±.011	.768±.007	.065±.014
MuSiQue	Original	.253±.018	.058±.011	.330±.007	.447±.002	.908±.003	.280±.021
MuSiQue	Randomized	.208±.018	.028±.011	.230±.035	.465±.008	.835±.003	.237±.018
MuSiQue	None	.212±.004	.000±.000	.015±.007	.048±.003	.124±.008	.028±.004
MuSiQue	Irrelevant	.205±.021	.015±.000	.048±.004	.306±.008	.674±.002	.003±.004
<i>Document-blind random-gate control (seed 0), kept fraction constant across all four modes</i>							
HotpotQA	All four modes			.465			
MuSiQue	All four modes			.465			

Table 1: Dual-arm randomization control. Doc. mode is the condition used for gate collection and training. Doc Acc./No-doc Acc. evaluate the updated student with original matched documents/without documents; *None* rows can therefore have nonzero Doc Acc. Keep, Agree, Conf., and Correct are pre-update gate retention, agreement, confidence, and correct-modal rate. Mean±std over two seeds.

5 Results

Document condition changes retention. On HotpotQA, retention falls from 0.380 ± 0.021 with the original document to 0.020 ± 0.000 with no document (Table 1). MuSiQue shows the same pattern under a leakage-resistant construction, falling from 0.330 ± 0.007 to 0.015 ± 0.007 . Randomized and irrelevant documents lie between the extremes on both arms, so the pattern occurs on both benchmarks. Figure 2 shows higher accuracy with evaluation-time document access in all four training modes, alongside decreasing retention across those modes.

Gate variants agree. Table 2 sweeps the original-vs.-none contrast across confidence-only, agreement-only, and joint gates. Removing the document sharply reduces all three on both arms, so the diagnostic is not tied to one parameterization.

Control and trace budget. The random gate uses a fixed keep probability independent of d , with the same seeded draws across modes; its realized fraction is 0.465. This checks a document-independent selection rule, not alternatives driven by confidence or domain familiarity. The retained-fraction ordering also holds across trace budgets (Table 3).

Interpretation and boundary. The tested gate changes substantially with document condition on both benchmarks. The diagnostic itself requires gate signals on four input forms, without retraining a student. The accompanying student accuracies describe the archived training runs; without an un-gated or untrained baseline under a matched budget,

they do not establish that greater document sensitivity improves downstream performance.

6 Discussion

Why the protocol composes with future gates.

The protocol needs only a gate signal for (q, d) on four input forms, so other gates can replace agreement or confidence while retaining the document-condition comparison. Each non-original mode adds a tutor-evaluation pass; student training is needed only to test downstream value.

Why selectivity is not enough. With fixed thresholds, lower agreement or confidence can reduce retention. The observed drop therefore cannot separate loss of answer support from domain familiarity or parseability. Matched counterfactual documents would test these alternatives.

What the diagnostic does not claim. The diagnostic does not prove that retained traces use correct evidence or that greater sensitivity improves the student; Table 1 does not isolate that causal relationship.

Gate variant	HotpotQA		MuSiQue	
	Orig.	None	Orig.	None
Confidence-only	.705	.020	.660	.020
Agreement-only	.485	.120	.410	.035
Joint (agr. + conf.)	.365	.020	.335	.020

Table 2: Gate-variant check at $M = 8$, seed 0 throughout. Joint values are the seed-0 component of Table 1.

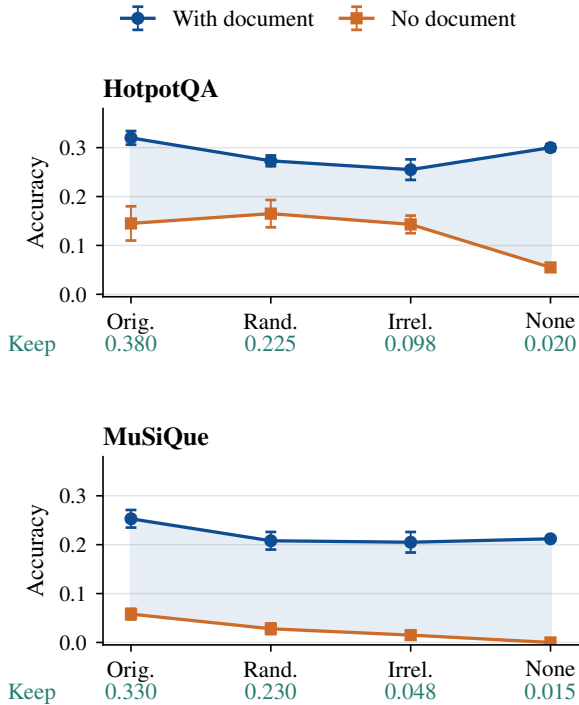


Figure 2: Student accuracy with original documents or without documents after training under each indicated document mode. Error bars show standard deviations over two seeds. Shading shows the evaluation-time document-access gap, not improvement over an untrained baseline; the bottom row reports gate retention K .

M	Original	Randomized	Irrelevant	None
4	.420	.238	.074	.016
8	.350	.214	.069	.019
16	.314	.243	.104	.018

Table 3: Trace-budget check: equal-weight means over both benchmarks and seeds 2 and 3 (48 archived runs; main-table seeds: 0 and 1).

Reporting recommendation. For grounded distillation papers, report the four-mode signature alongside parse coverage and task metrics. Let S_{orig} and S_{none} denote the retained question sets. Their overlap $|S_{\text{orig}} \cap S_{\text{none}}|/|S_{\text{orig}}|$ measures how much original-mode supervision also passes without documents. The tutor’s no-document correctness on S_{orig} tests memory-only answering in that cohort; it is distinct from the student’s validation accuracy in Table 1. Equal retention rates can describe disjoint sets. The archived summaries lack these question-level records, so neither cohort statistic nor the all-unparseable rate can be recovered. Use matched seeds and question IDs for paired comparisons; mode-wise averages discard the within-question relation.

Why this matters for supervision quality. Retained trajectories become examples that shape a student. A gate that selects memory-answerable questions may transmit shortcuts, even when its aggregate retention is document-sensitive. The four-mode comparison checks the supervision source before downstream training, while cohort overlap and evidence attribution would determine which retained questions drive the signature. These are complementary checks on the amount and identity of selected supervision.

Failure patterns exposed by the four modes. High no-document retention is compatible with parametric confidence, while high randomized or irrelevant retention can reflect domain or surface effects. The observed ordering shows sensitivity to document condition. Because the interventions jointly alter support, domain, lexical overlap, and prompt naturalness, they do not uniquely isolate evidence use. Same-benchmark random documents retain more questions than cross-benchmark documents on both datasets, although neither is matched to the question. Correct-modal rate need not follow retention: on HotpotQA it is 0.323 under randomized documents versus 0.287 under original documents. Retention therefore cannot substitute for checking correctness or evidence support.

Relation to final-answer grounding. Many grounding evaluations ask whether a final answer is supported by a passage. Our setting is earlier in the pipeline: the gate decides which tutor trajectories become training data before the student is evaluated. The proposed contrast complements answer-level faithfulness checks by testing whether the supervision filter responds to document condition.

7 Conclusion

We introduced a four-mode randomization-control diagnostic for asking whether self-distillation gates in grounded QA depend on the supplied document. Applied to HotpotQA and MuSiQue, the joint agreement-confidence gate responds strongly to document condition, while a document-blind control remains flat.

The four-mode contrast audits document sensitivity alongside accuracy and confidence. It does not establish evidence attribution or downstream training benefits.

Limitations

Our experiments cover one tutor model and two QA benchmarks, which limits the generality of our findings. Future work should evaluate the diagnostic across broader models, datasets, and domains.

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